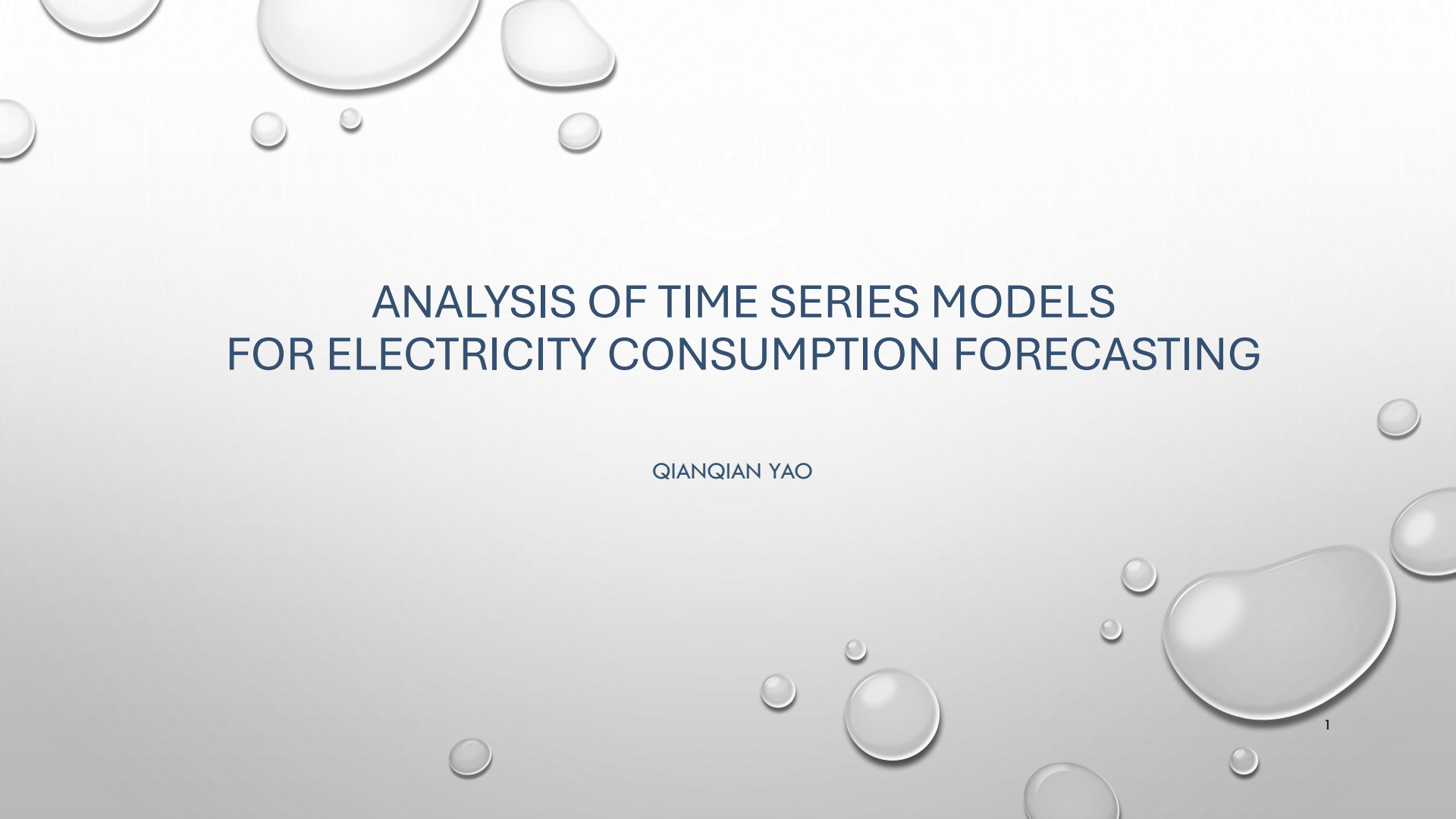


The background of the slide is a light gray gradient. It is decorated with numerous realistic water droplets of various sizes. Some droplets are at the top left, some are scattered in the middle, and a larger cluster is at the bottom right. Each droplet has a highlight and a shadow, giving it a 3D effect.

ANALYSIS OF TIME SERIES MODELS FOR ELECTRICITY CONSUMPTION FORECASTING

QIANQIAN YAO

CONTENTS

- BACKGROUND OF TIME SERIES ANALYSIS
- PROJECT MOTIVATION
- DATA
- MODELS
- CONCLUSION

A **Time Series** is a collection of observations x_t , each of which is indexed by time t , ie., the observation x_t is made at a specific time t .

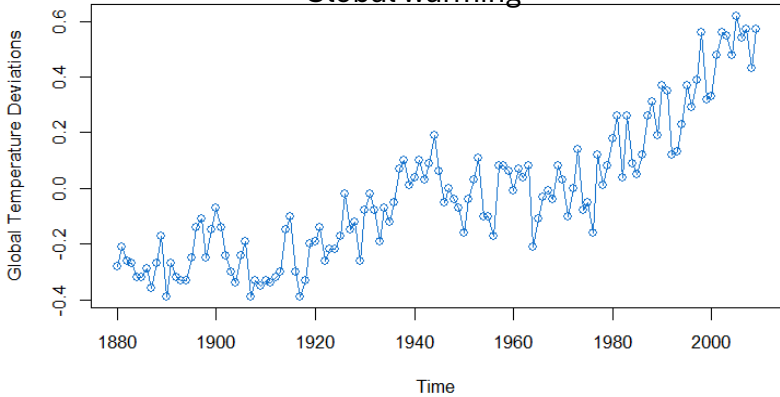
A **Time Series Model** for the observed data $\{x_t\}$ is a specification of the joint distributions (or possibly only the mean and covariance) of a sequence of random variables $\{X_t\}$ of which $\{x_t\}$ is postulated to be a realization.

Objectives of Time Series Analysis: Modeling and Forecasting

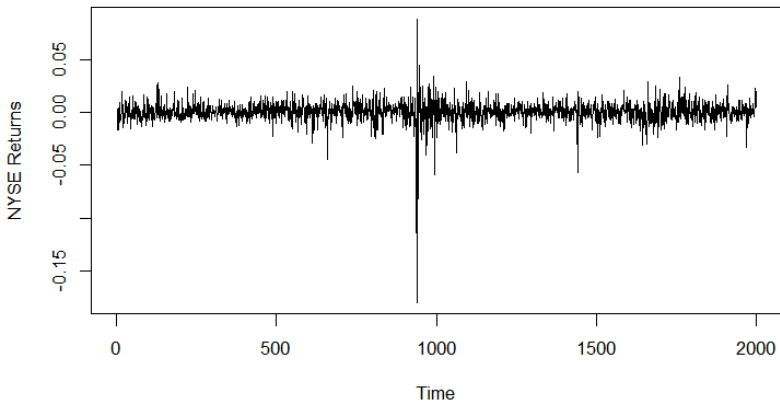
- 1- set up a probability model to represent data;
- 2- estimate parameters in the model;
- 3- check the goodness of the model fit for the data;
- 4- use the fitted model to enhance our understanding of the mechanism generating the series;
- 5- once the probability model has been established, we may use it to summarize the data or forecast (predict) the future values, etc.

Real World Examples

Global warming

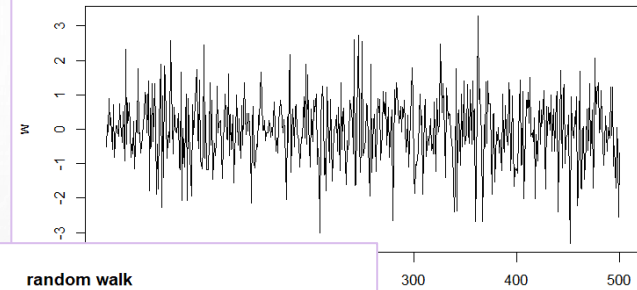


New Yorker Stock Exchange

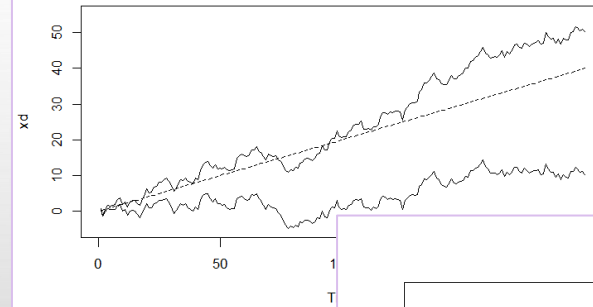


Model Generated Examples

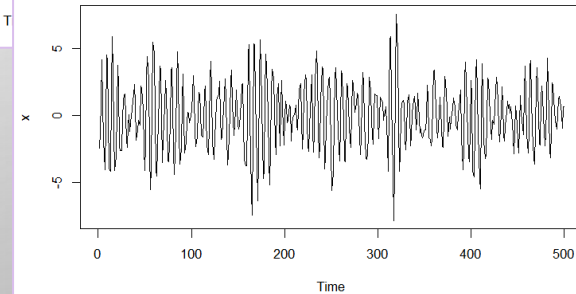
white noise



random walk



autoregression



Stationary

$\{X_t\}$ is stationary if

- 1) $\mu_t = E(X_t)$ is independent of t (it is a constant);
- 2) $\text{Cor}(X_t, X_{t+h})$ is a function of h only;

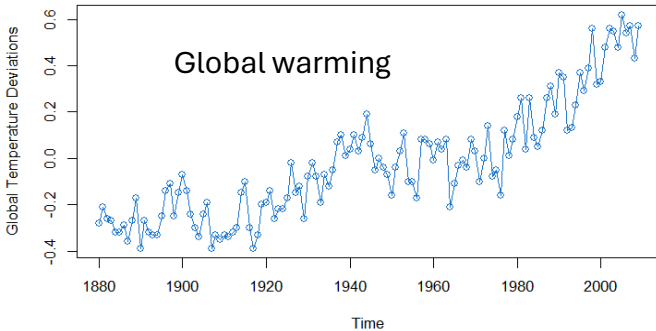
This condition makes sure the data are homogeneous, so that the data can be modeled.

The ways to check stationary: *Dickey Fuller*, *ACF* & *PACF*

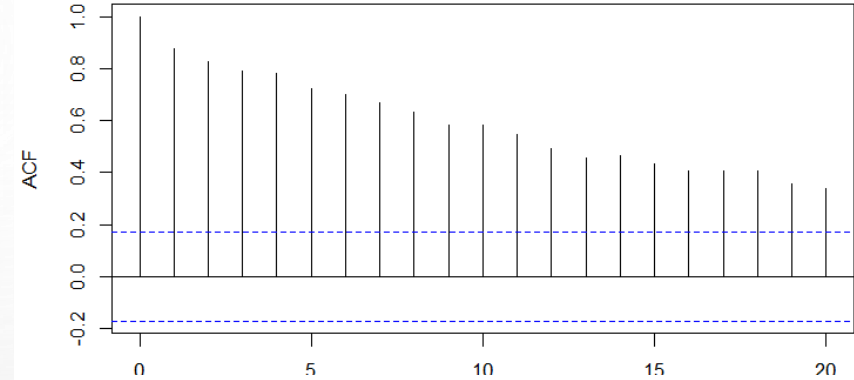
```
{r}
adf.test(gtemp)
```

Augmented Dickey-Fuller Test

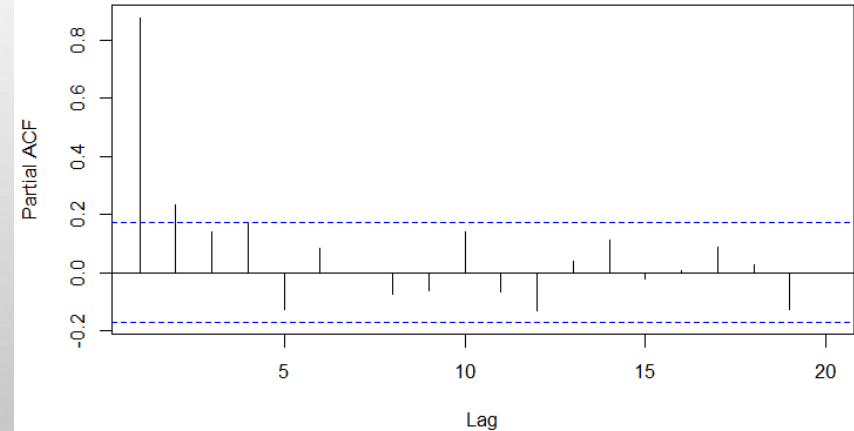
```
data: gtemp
Dickey-Fuller = -1.6893, Lag order = 5, p-value = 0.7057
alternative hypothesis: stationary
```



Series gtemp



Series gtemp

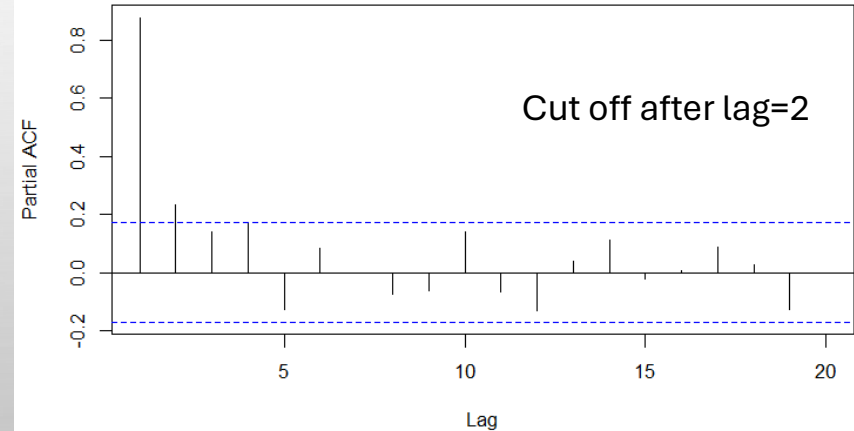
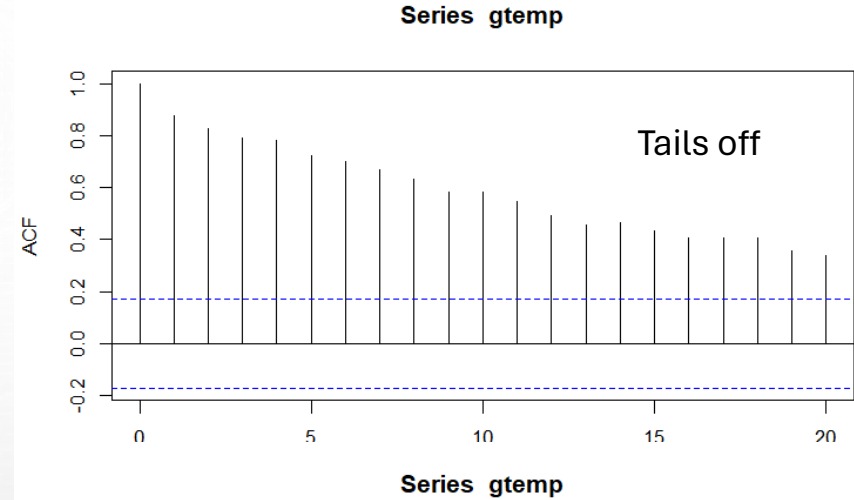


Summary: Behavior of ACF and PACF for ARMA Models

	AR(p)	MA(q)	ARMA(p,q)
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

Modeling selection for Global Warming: AR(2)

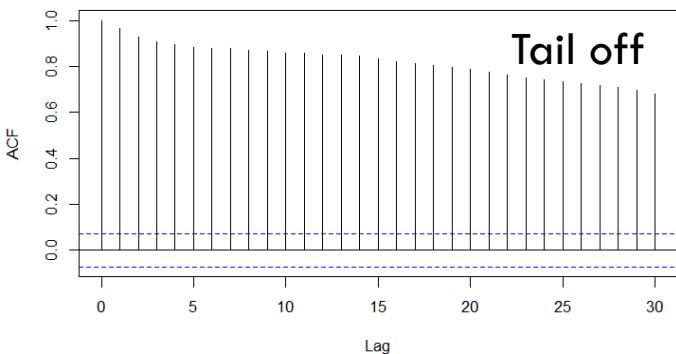
We are not sure whether there are other models appropriate for Global Warming



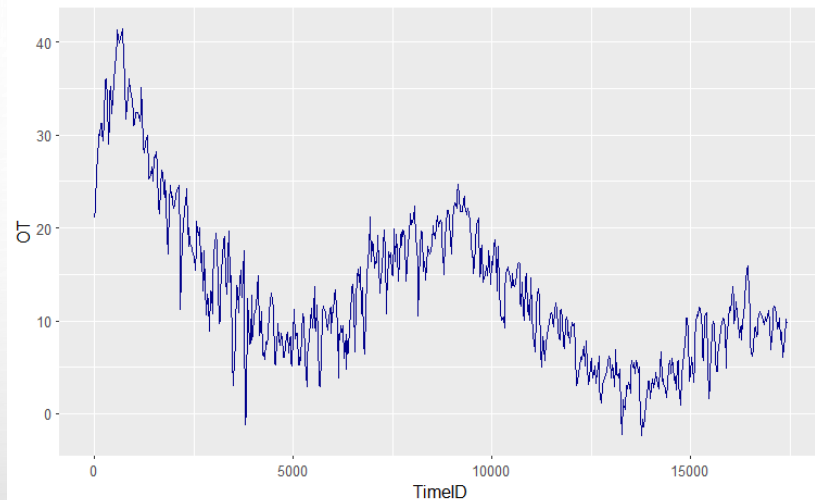
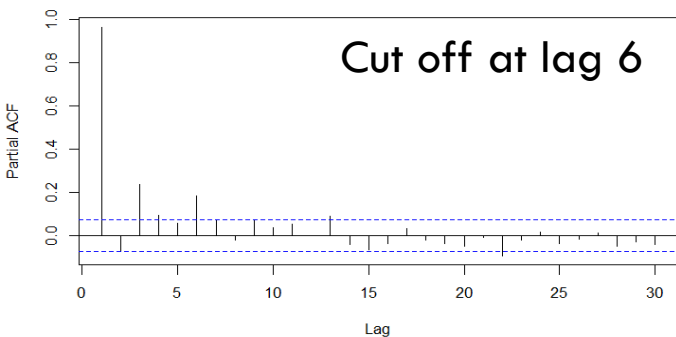
- To address climate change, we can discover catalysts to develop new energy resources, on the other hand we also should save the current energy resources.
- **Predicting the electric power consumption precisely on demand can save energy.**
- The electric power distribution problem is the distribution of electricity to different areas depends on its sequential usage, but predicting the following demand of a specific area is difficult, as it varies with weekdays, holidays, seasons, weather, temperatures, etc.
- It causes unnecessary waste of electricity and equipment depreciation.
- There is one efficient strategy to solve this problem is **to predict the electrical transformers' oil temperature because the oil temperature can reflect the electricity transformer. So that we can predict electric power consumption precisely.**
- The **dataset** is historical records of oil temperatures measured at electricity distribution stations.
- This **project** focuses on time series data analysis, forecasting model selection and evaluation.

Time	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
Days	High Useful Load	High Useless Load	Middle Useful Load	Middle Useless Load	Low Useful Load	Low Useless Load	Oil Temperature

Series OT



Series OT

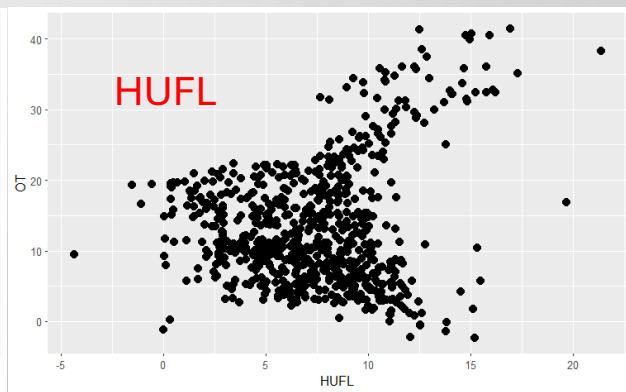
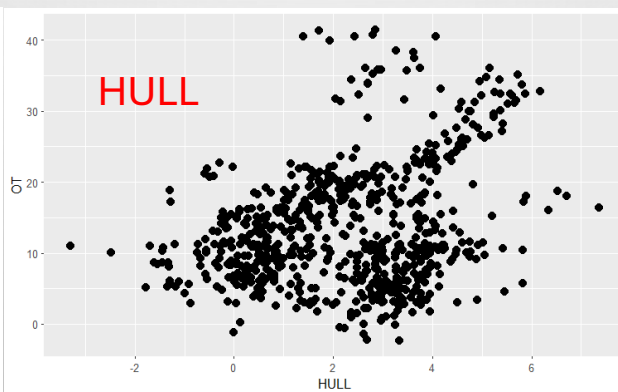
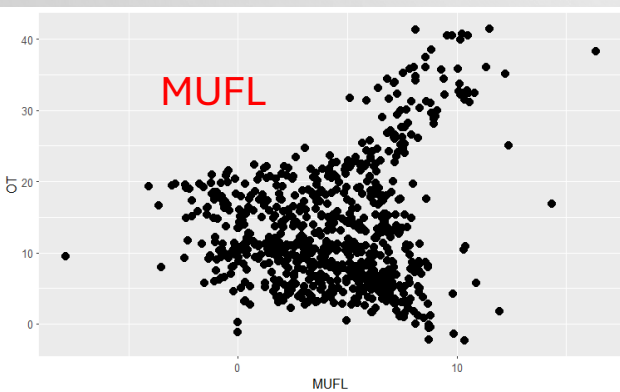
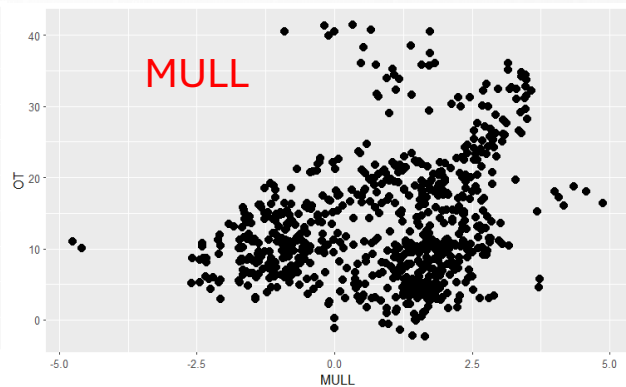
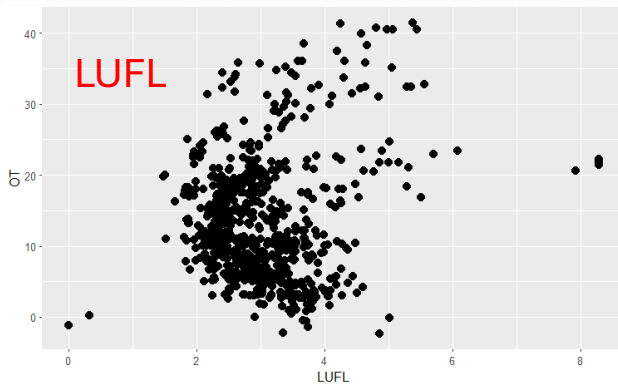
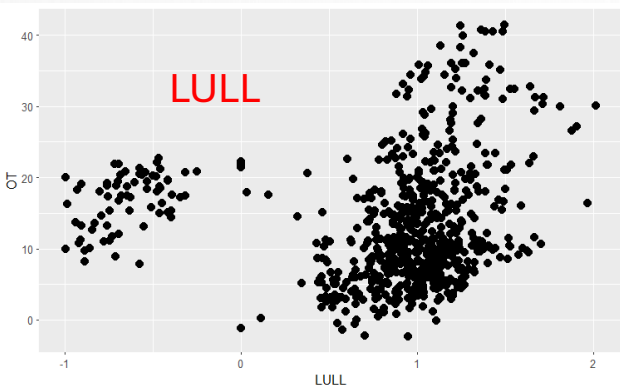


```
{r}
adf.test(OT)
```

Augmented Dickey-Fuller Test

```
data: OT
Dickey-Fuller = -2.137, Lag order = 8, p-value = 0.5203
alternative hypothesis: stationary
```

Time	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
Days	High	High	Middle	Middle	Low	Low	Oil
	Useful	Useless	Useful	Useless	Useful	Useless	Temperature
	Load	Load	Load	Load	Load	Load	



polynomial regression at degree 2

```
call:
lm(formula = OT ~ trend + huf1 + hull + muf1 + mull + (luf12) +
    (lull2))
```

Residuals:

	Min	1Q	Median	3Q	Max
	-17.4020	-3.6848	0.1798	3.3267	16.1758

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	23.002979	0.491293	46.821	< 2e-16 ***
trend	-0.031206	0.001091	-28.615	< 2e-16 ***
huf1	1.926506	0.361013	5.336	1.27e-07 ***
hull	4.798562	0.642591	7.468	2.37e-13 ***
muf1	-2.591358	0.386440	-6.706	4.06e-11 ***
mull	-3.809663	0.692627	-5.500	5.28e-08 ***
luf12	0.622398	0.076995	8.084	2.67e-15 ***
lull2	3.805969	0.523819	7.266	9.69e-13 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

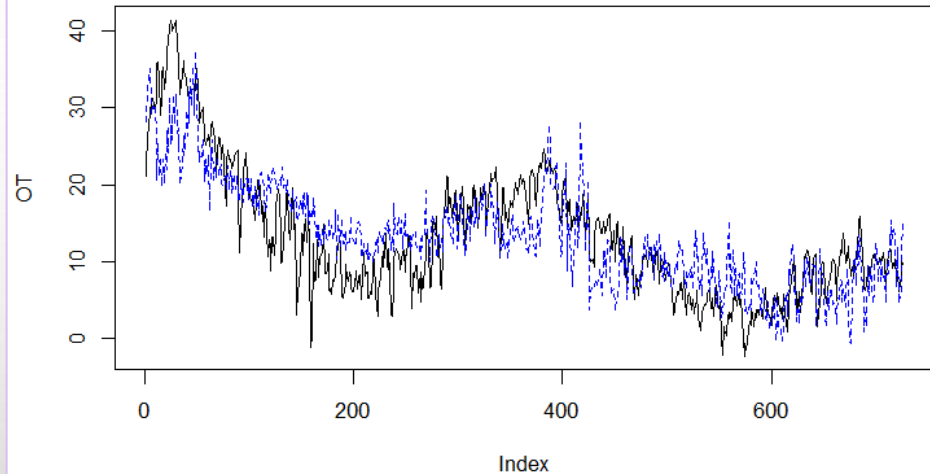
```
Residual standard error: 5.088 on 718 degrees of freedom
Multiple R-squared:  0.6377,    Adjusted R-squared:  0.6341
F-statistic: 180.5 on 7 and 718 DF,  p-value: < 2.2e-16
```

Analysis of Variance Table

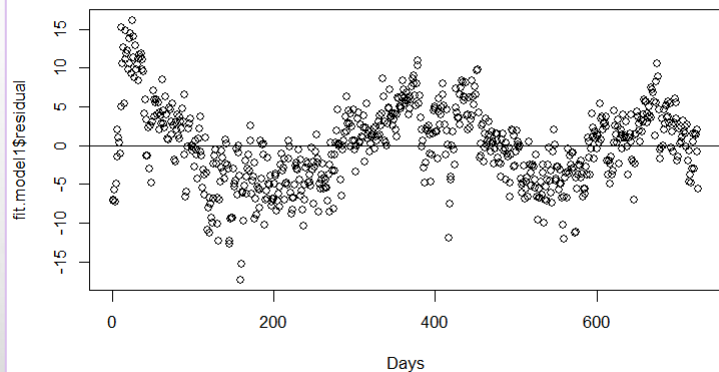
Response: OT

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
trend	1	20715.3	20715.3	800.1704	< 2.2e-16 ***
huf1	1	1.3	1.3	0.0502	0.8228
hull	1	5564.1	5564.1	214.9251	< 2.2e-16 ***
muf1	1	2801.9	2801.9	108.2308	< 2.2e-16 ***
mull	1	25.8	25.8	0.9949	0.3189
luf12	1	2236.1	2236.1	86.3729	< 2.2e-16 ***
lull2	1	1366.7	1366.7	52.7920	9.689e-13 ***
Residuals	718	18588.0	25.9		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Check Residual IID



Box-Pierce test

```
data: fit.model1$residuals
X-squared = 6352.6, df = 38, p-value < 2.2e-16
```

Not IID

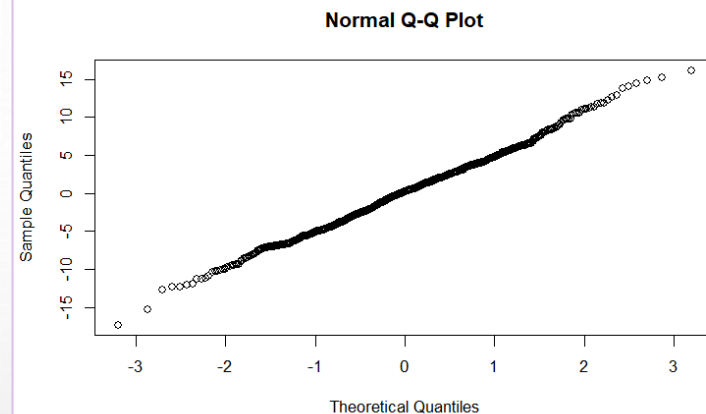
Box-Pierce test

```
data: fit.model1$residuals
X-squared = 6352.6, df = 39, p-value < 2.2e-16
```

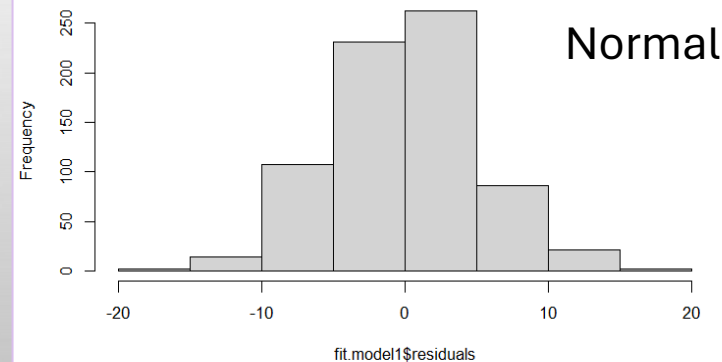
Box-Pierce test

```
data: fit.model1$residuals
X-squared = 6352.6, df = 40, p-value < 2.2e-16
```

Check Residual Normality



Histogram of fit.model1\$residuals



```
fit.model1 <- lm(OT ~ trend + hufl + hull + mufl + mull + (lufl2) + (lull2))
```

```
fit.model2 <- lm(OT ~ trend + hufl + hull + mufl + mull + (lufl +lufl2) + (lull +lull2))
```

```
fit.model3 <- lm(OT ~ trend + (hufl+hufl2) + (hull+hull2) + (mufl+mufl2) + (mull+mull2) + (lufl+lufl2) + (lull+lull2))
```

Model	AIC	BIC	AICc
Model_1	4.2675	4.3244	4.2594
Model_2	4.2669	4.3364	4.2533
Model_3	4.2515	4.3463	4.2269

Time series of Oil Temperature

The first step in the analysis of any time series is to plot the data. It is advisable that

1. if there are any apparent discontinuities in the series, such as a sudden change of level, it may be advisable to analyze the series by first breaking it into homogeneous segments;
2. if there are outlying observations, they should be studied carefully to check whether there is any justification for discarding them (as for example if an observation has been incorrectly recorded).

The line plot of a time series may also suggest the possibility of modeling the data by the *classical decomposition model*.

$$X_t = m_t + s_t + Y_t, \quad (\text{Model 1.5.1 in reference book, BD})$$

where m_t is the trend, s_t is the seasonality with periodicity d and Y_t is the noise (random and stationary).

Steps in ARIMA modeling

Step1: differencing to get Y_t

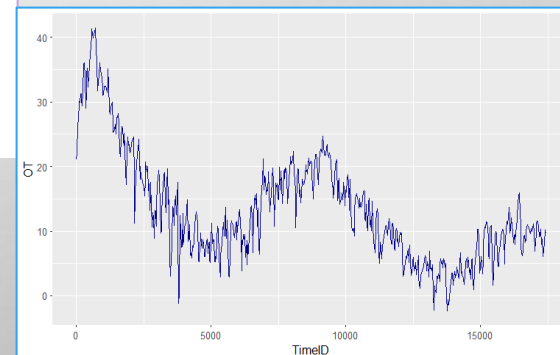
Step2: check ACF/ PACF on Y_t

Step3: Fit AR(p)

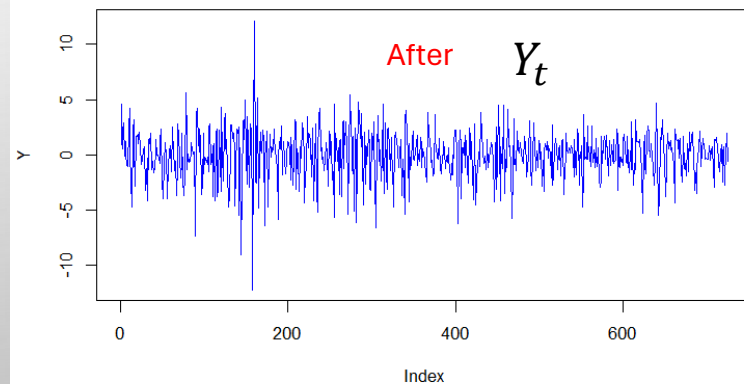
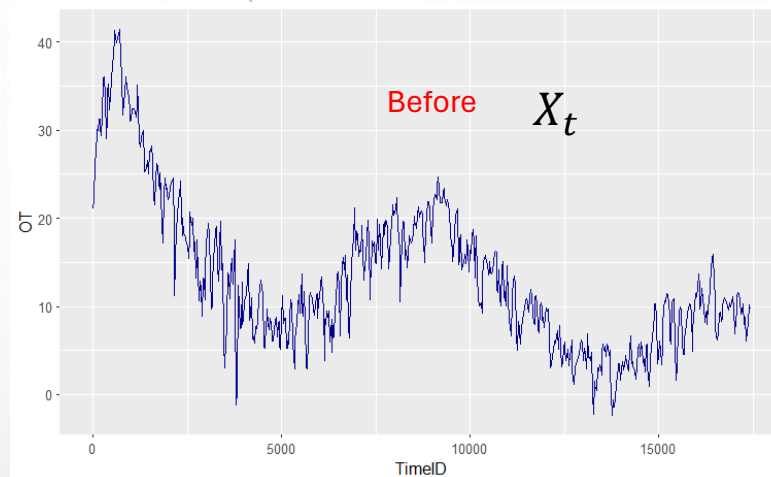
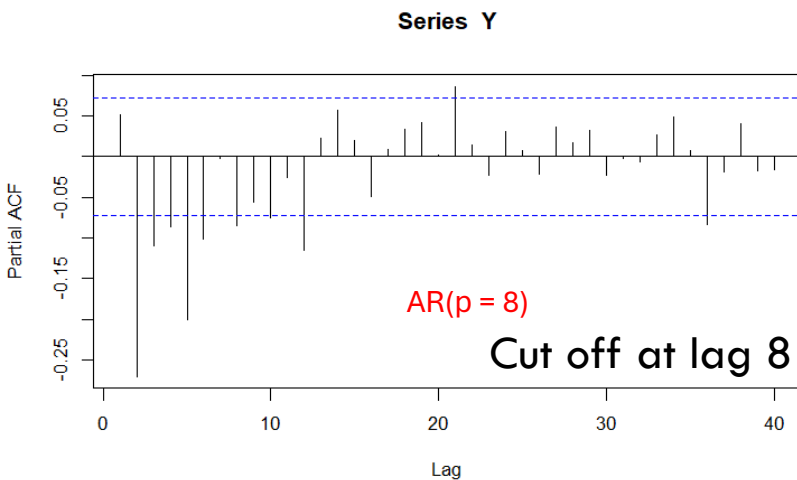
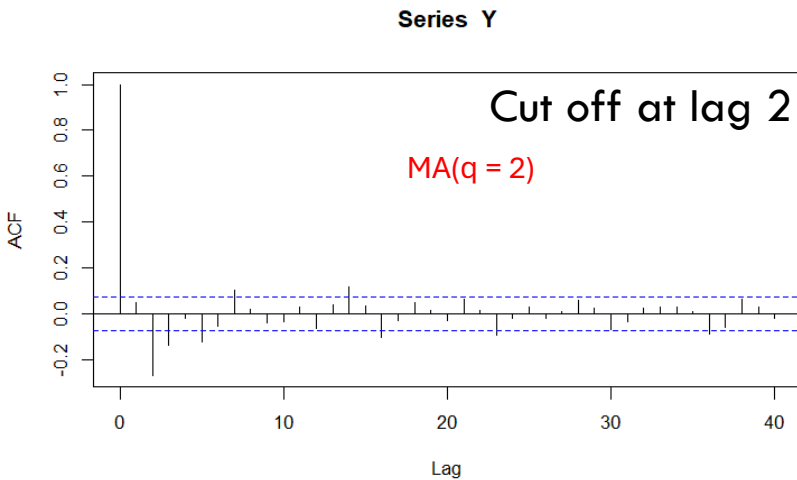
Step4: Fit MA(q)

Step5: Fit ARIMA(p, d, q)

Step6: What If Fit X_t directly?



differencing to get $Y_t = X_t - X_{t-1}$



AR(p = 8)

model.fit1 <- arima(Y, order=c(8, 0, 0))

Call:

arima(x = pred, order = c(8, 0, 0))

Coefficients:

	ar1	ar2	ar3	ar4	ar5	ar6	ar7	ar8
	-0.016	-0.3323	-0.2058	-0.1319	-0.2228	-0.1365	-0.0100	-0.0900
s.e.	0.037	0.0372	0.0389	0.0389	0.0388	0.0390	0.0372	0.0372
intercept								
	-0.0226							
s.e.	0.0353							

sigma^2 estimated as 4.124: log likelihood = -1542.64, aic = 3105.29

Box-Pierce test

data: model.fit1\$residuals
X-squared = 45.047, df = 38, p-value = 0.2008

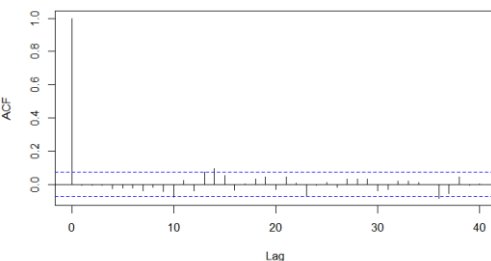
Box-Pierce test

data: model.fit1\$residuals
X-squared = 45.047, df = 39, p-value = 0.2336

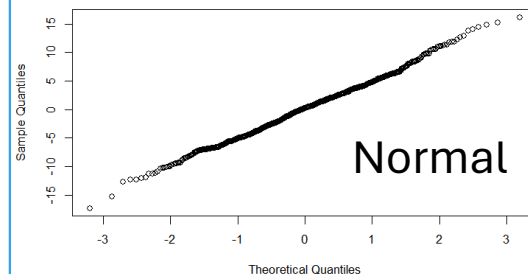
Box-Pierce test

data: model.fit1\$residuals
X-squared = 45.047, df = 40, p-value = 0.2689

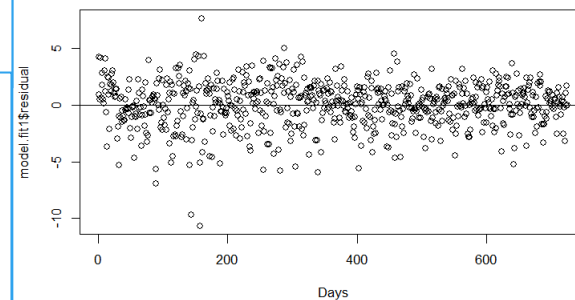
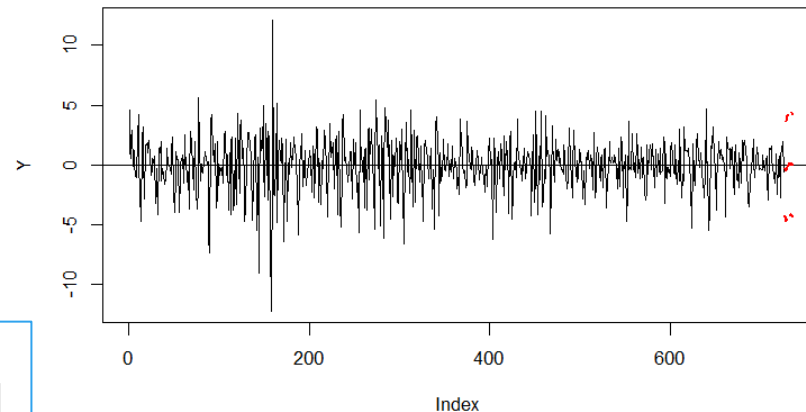
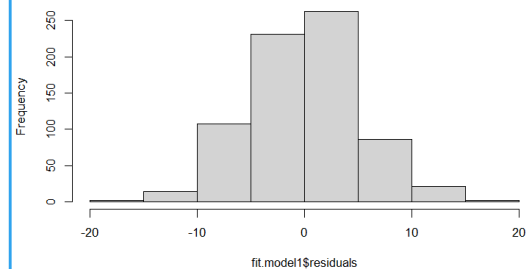
Series model.fit1\$residuals



Normal Q-Q Plot



Histogram of fit.model1\$residuals



MA(q = 2)

```
model.fit2 <- arima(Y, order=c(0, 0, 2))
```

Call:

```
arima(x = Y, order = c(0, 0, 2))
```

Coefficients:

	ma1	ma2	intercept
	-0.1088	-0.3928	-0.0215
s.e.	0.0403	0.0433	0.0389

sigma² estimated as 4.393: log likelihood = -1565.39, aic = 3138.77

Box-Pierce test

data: model.fit2\$residuals
X-squared = 87.758, df = 38, p-value = 8.258e-06

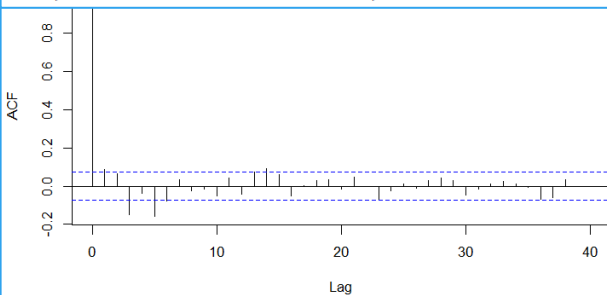
Box-Pierce test

Not IID

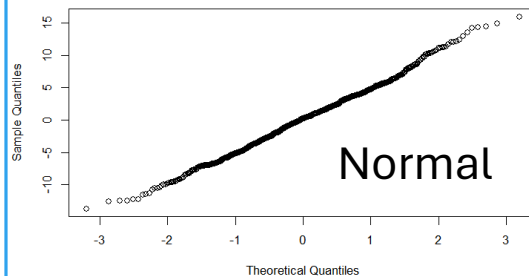
data: model.fit2\$residuals
X-squared = 87.758, df = 39, p-value = 1.286e-05

Box-Pierce test

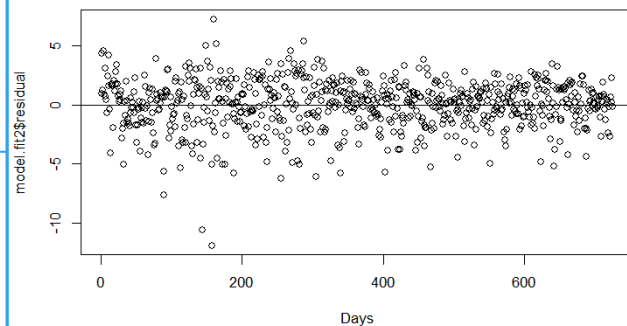
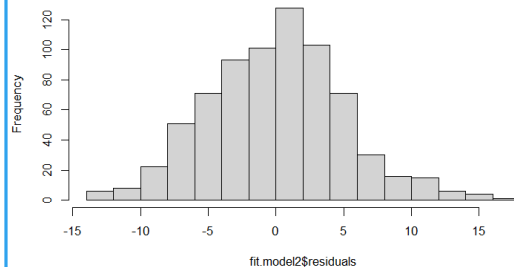
data: model.fit2\$residuals
X-squared = 87.758, df = 40, p-value = 1.978e-05



Normal Q-Q Plot



Histogram of fit.model2\$residuals



ARIMA(p, d, q)

```
auto <- auto.arima(Y, d=0, ic="aicc", trace=T)
```

Series: Y
ARIMA(2,0,2) with zero mean

Coefficients:
ar1 ar2 ma1 ma2
0.5770 -0.1357 -0.6095 -0.1958
s.e. 0.1197 0.0938 0.1183 0.1076

sigma^2 = 4.188: log likelihood = -1546.22
AIC=3102.44 AICc=3102.52 BIC=3125.37

Box-Pierce test

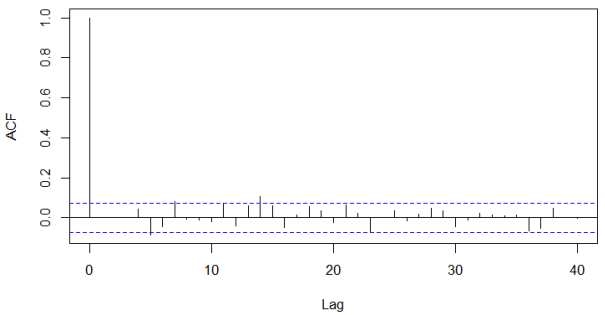
data: auto\$residuals
X-squared = 56.399, df = 38, p-value = 0.02767

Box-Pierce test

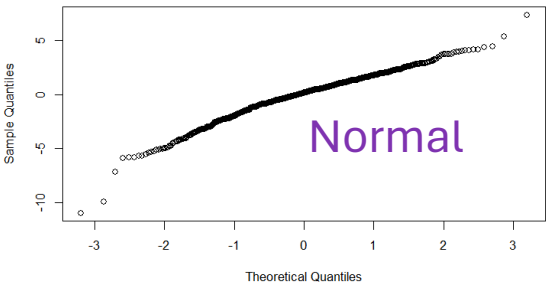
data: auto\$residuals
X-squared = 56.399, df = 39, p-value = 0.03525

IID

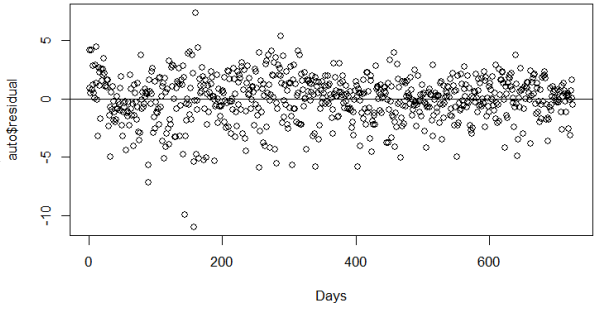
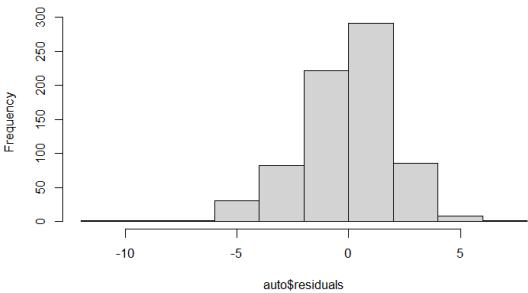
Series auto\$residuals



Normal Q-Q Plot



Histogram of auto\$residuals



ARIMA(p, d, q) on X_t OT.fit1 <- **auto.arma**(OT, trace=T, ic="aicc")

Series: OT
ARIMA(2,1,2)

Coefficients:

	ar1	ar2	ma1	ma2
	0.5770	-0.1357	-0.6095	-0.1958
s.e.	0.1197	0.0938	0.1183	0.1076

sigma^2 = 4.188: log likelihood = -1546.22
AIC=3102.44 AICc=3102.53 BIC=3125.37

Box-Pierce test

data: OT.fit1\$residuals
X-squared = 56.483, df = 38, p-value = 0.0272

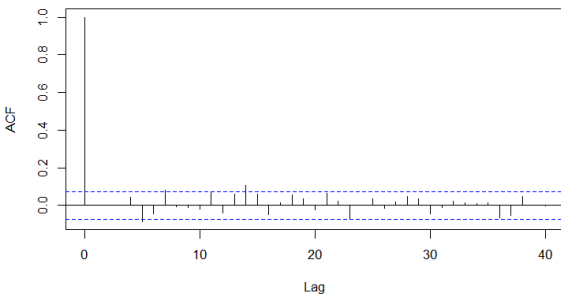
Box-Pierce test

data: OT.fit1\$residuals
X-squared = 56.483, df = 39, p-value = 0.03467

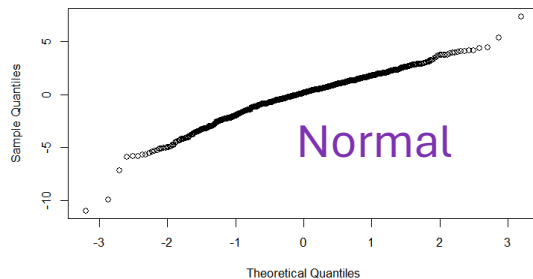
Box-Pierce test

IID

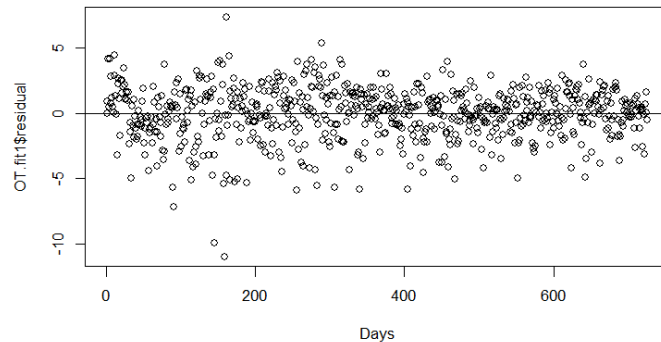
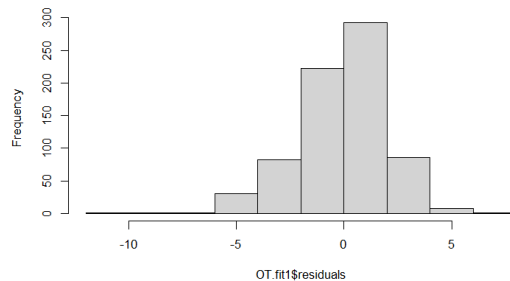
Series OT.fit1\$residuals



Normal Q-Q Plot



Histogram of OT.fit1\$residuals



```

model.fit1 <- arima(Y, order=c(8, 0, 0))
model.fit2 <- arima(Y, order=c(0, 0, 2))
auto <- auto.arima(Y, d=0, ic="aicc", trace=T)
auto2 <- auto.arima(Y, ic="aicc", trace=T)
OT.fit1 <- auto.arima(OT, trace=T, ic="aicc")

```

```

num <- length(OT)
# AIC
AIC(OT.fit1)/num - log(2*pi)
# BIC
AIC(OT.fit1, k=log(num))/num - log(2*pi)
# AICc
log(sum(resid(OT.fit1)^2)/num) + (num + 5)/(num - 5 - 2)

```

Model	AIC	BIC	AICc
AR(p = 8) on Y_t	2.4394	2.5026	2.4322
MA(q = 2) on Y_t	2.4855	2.5108	2.4952
ARIMA(2,0,2) on Y_t	2.4354	2.4670	2.4421
ARIMA(2,1,2) on X_t	2.4354	2.4670	2.4421

Linear Regression Models

Residual Assumption of IID is not satisfied, add more feature couldn't improve the model performance obviously, so linear regression model may not a good choice.

ARIMA

Among AR, MA, ARIMA, ARIMA model performs better.

Process on decomposed Y_t : ARIMA(2, 0, 2) is the best choice.

Process on Original X_t : ARIMA(2, 1, 2) is the best choice.

Which make sense, because $d=1$ is the “on step differencing” from X_t to get Y_t

More details will be covered in final report

- Causality, invertibility,
- ARFIMA (long memory process)
- Check whether ARCH, GARCH model are applicable

THANKS

Q & A